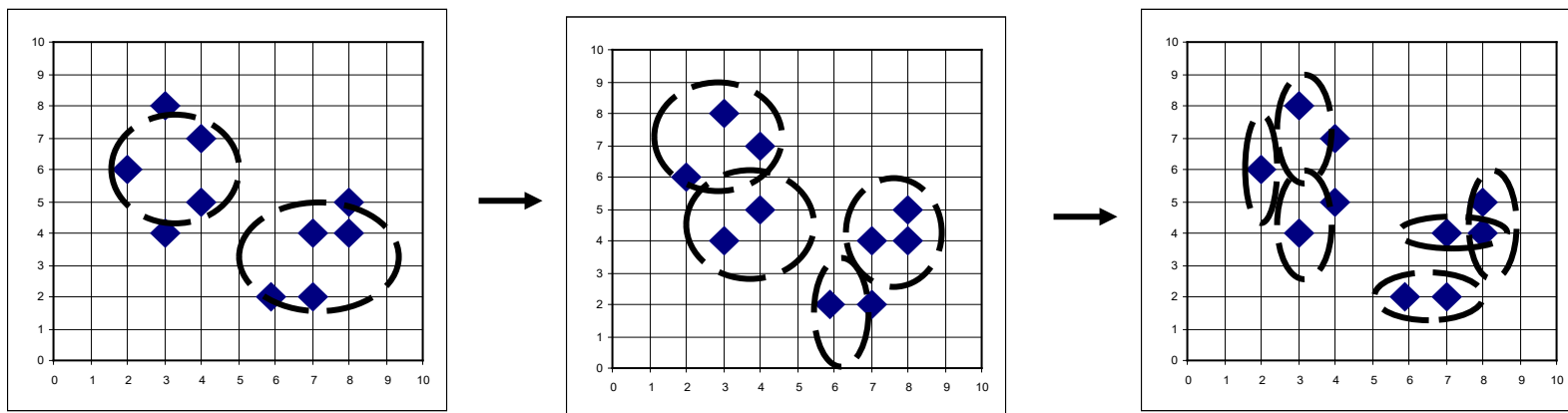


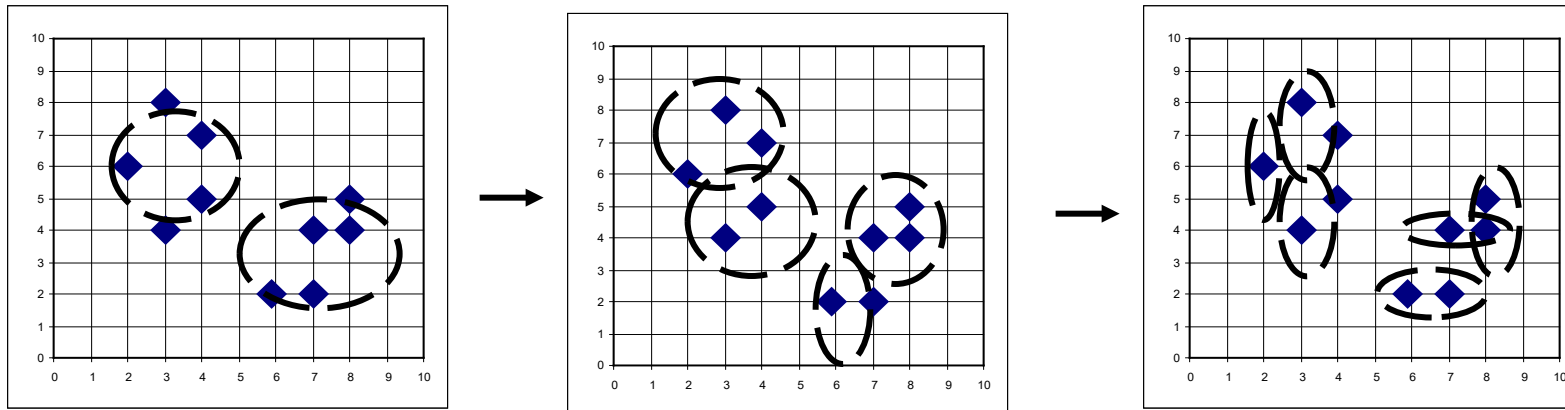
Divisive Clustering

- ❑ DIANA (Divisive Analysis) (Kaufmann and Rousseeuw, 1990)
 - ❑ Implemented in some statistical analysis packages, e.g., Splus
- ❑ Inverse order of AGNES: Eventually each node forms a cluster on its own



Divisive Clustering Is a Top-down Approach

- ❑ The process starts at the root with all the points as one cluster
- ❑ It recursively splits the higher level clusters to build the dendrogram
- ❑ Can be considered as a global approach
- ❑ More efficient when compared with agglomerative clustering



More on Algorithm Design for Divisive Clustering

- ❑ Choosing which cluster to split
 - ❑ Check the sums of squared errors of the clusters and choose the one with the largest value
- ❑ Splitting criterion: Determining how to split
 - ❑ One may use Ward's criterion to chase for greater reduction in the difference in the SSE criterion as a result of a split
 - ❑ For categorical data, Gini-index can be used
- ❑ Handling the noise
 - ❑ Use a threshold to determine the termination criterion (do not generate clusters that are too small because they contain mainly noises)